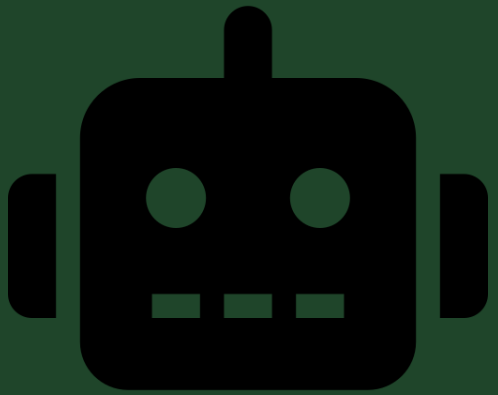
GRUPO 4

GARDENING ROBOT



Solugarden – Advanced Manufacturing Technology

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FATEC São Bernardo do Campo | June 2026

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SOLUGARDEN

About the Company

Fictional company created to integrate knowledge from the Advanced Manufacturing program at FATEC SBC. Develops autonomous gardening robots for green area maintenance in condominiums and commercial spaces.

Mission

Technological solutions for automation of green areas, promoting efficiency and safety.

Objective

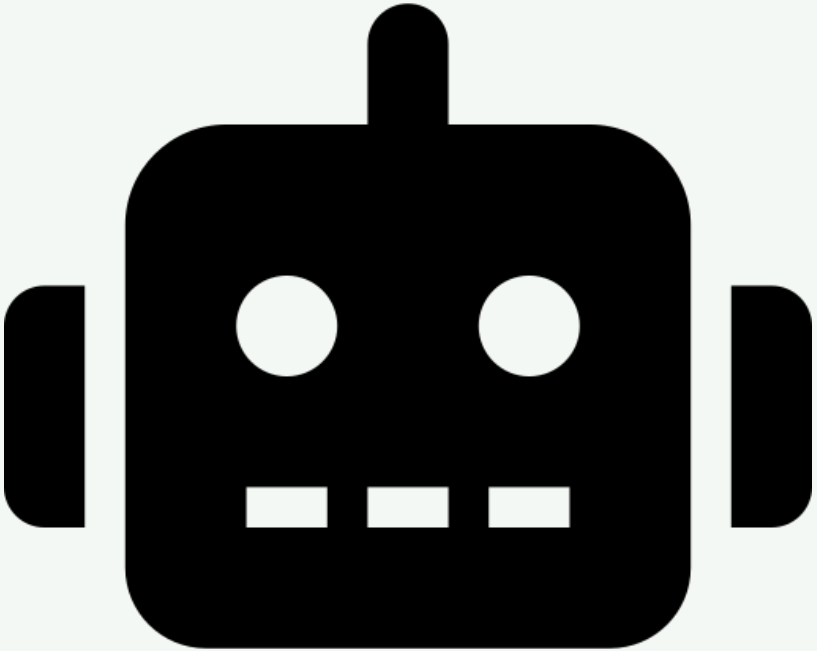
Autonomous robots that mow with precision, reducing manual labor and occupational risks.

Application

Residential condominiums – with projected expansion to industrial scale.

Differentials

100% electric delivery vehicles; real-time data via Azure cloud.



PRODUCT DESCRIPTION

Chassis & Drive

Metal + ABS/PA thermoplastic. Two DC drive wheels with gearbox + free wheel. Supports up to 30–40% inclines.

Electronics

Arduino/Raspberry Pi MCU. H-bridge motor drivers. Bluetooth/Wi-Fi. OTA firmware updates.

Cutting System

Rotary discs with hardened steel blades. Height 20–50 mm, bidirectional. Cutting width: 22 cm.

Sensors

Ultrasonic + IR obstacle detection, cliff sensors, tilt sensor, front bumper + perimeter boundary wire.

Battery & Performance

60 min cutting / ~60 min recharge. Li-Ion 20V 5Ah. Cut precision >90%. Weight ~10 kg.

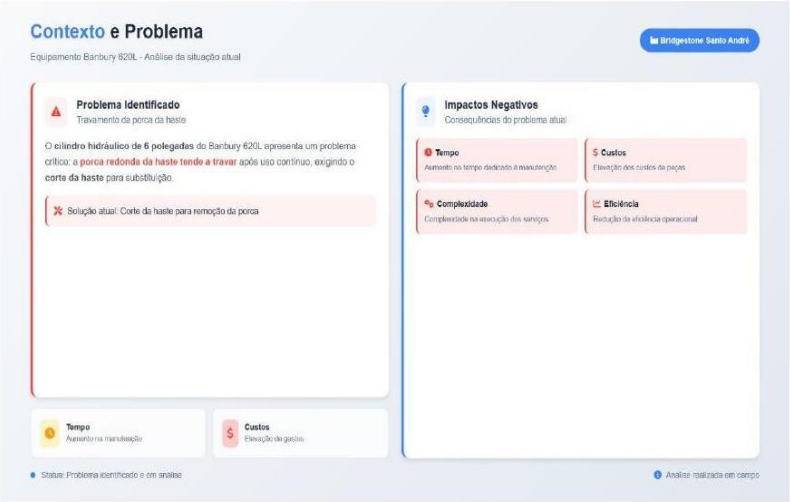
Software

Probabilistic navigation, spiral sweep, terrain mapping via odometry. App control + scheduling.

DESIGN THINKING – FACTORY PROJECT

Applied to reduce corrective maintenance time on hydraulic equipment at Solugarden's partner supplier.

Figura 1 - Contexto e problema



Fonte: Autoria própria, 2026

1

Empathy

Real maintenance observation; physical wear, risks, unproductive downtime.

2

Definition

Progressive jamming of hydraulic rod locking nut identified as root cause.

3

Ideation

Split block + permanent fixed nut on rod.

4

Prototyping

CNC-machined Steel 1045 components.

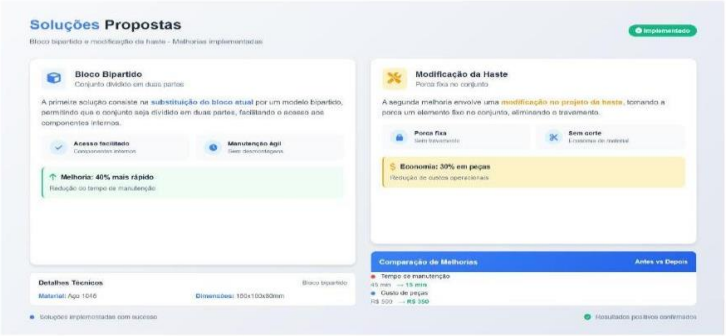
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Testing

Reduced time, safer disassembly, no rod cutting needed.

RESULTS – DESIGN THINKING

Figura 3: Soluções Propostas



Fonte: Autoria própria, 2026

↓ 60% Maintenance Time Reduction

↓ 30% Component Replacement Cost

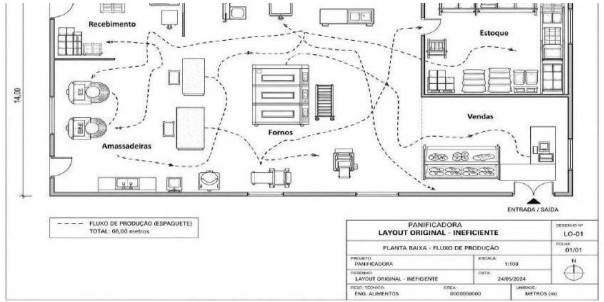
↑ 95% Operational Reliability

↑ + Equipment Availability & Supply Continuity

LAYOUT OPTIMIZATION – BAKERY

Company X – Campinas/SP | 336 m² | 18 employees | Revenue: R$ 120,000/month | ~140 active SKUs

BEFORE – Spaghetti Flow



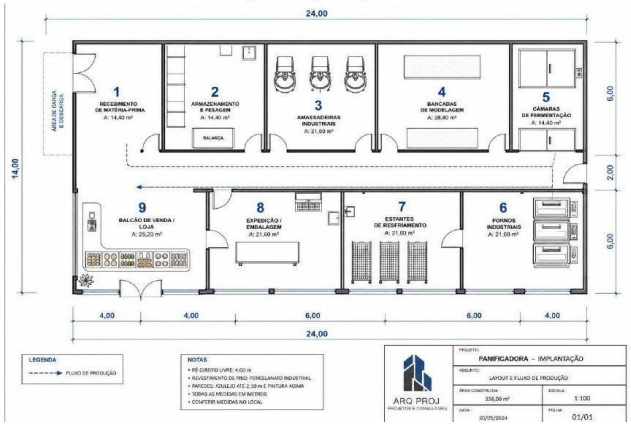
Fonte: Autoria própria, 2026

Alem disso, conforme tabela 2, podemos verificar alguns desperdicios identificados no layout atual.

Tabela 2 - Matriz de Desperdicios

AFTER – U-Shaped Cell Layout

Figura 9 - Layout Proposto



Wastes Identified: Transport (contamination risk) • Waiting (cooling shelves) • Movement (long distances) • Defects (12.4% dough loss)

Standards: ANVISA RDC 2016/2004 | NR-12 (Machine Safety) | NR-17 (Ergonomics)

RESULTS – U-SHAPED LAYOUT

Indicator (KPI)	Before	After	Gain
Distance per Batch	68 m	24 m	**–64.7%**
Lead Time	260 min	158 min	**–39.2%**
Loss / Defect Index	12.4%	< 5.0%	**–59.6%**
OEE (Overall Efficiency)	62%	80%	**+29.0%**
Monthly Inventory Turnover	2.3×	5.0×	**+117.4%**

U-shaped rearrangement + Lean practices → agility during demand peaks and reduced operator fatigue.

TEAM MANAGEMENT

Organizational structure of Solugarden focused on leadership, SMART goals and people development.

Org Chart

Distribution of roles and responsibilities by department.
Clear hierarchy between board, managers, and employees.

SMART Goals

Defined by sector/department.
Deadlines, owners, and priority levels established.



HR Indicators

Absenteeism (target: <4%)
Turnover (target: <10% p.a.)
GPTW Trust Index (target: ≥70%)

Performance Evaluation

Scale 1–5 by behavioral competencies.
Individual Development Plan (PDI) with clear goals and deadlines.

PERSONAL BRANDING

Survey of the most highlighted competencies and values by group members throughout the Solugarden project.



These competencies were identified through keyword analysis across group members' self-assessments during the Solugarden project development.

AUTOMATED SYSTEMS II – ELEVATED CONVEYOR



Fonte: ADZ máquinas e equipamentos, 2026

Source: ADZ Máquinas e Equipamentos, 2026

System Specifications

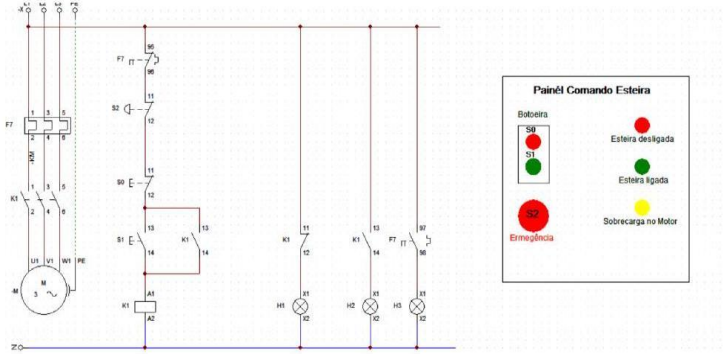
Length:	2,744 mm
Max. Elevation:	1,840 mm
Operating Pressure:	6 bar (0.6 MPa)
Cylinders:	2× SMC CDA2D100-600Z (double-action)
Valve:	5/2-way, dual solenoid – SY5120-5DZ
Control:	PLC 24VCC (Schneider TM172)
Force – Advance:	4,712.4 N
Force – Retract:	4,417.9 N

AUTOMATED SYSTEMS II – DIAGRAMS & PLC

Electrical Control Diagram (CadSimu)

ligada, e por fim H3 representando uma sobrecarga no motor.

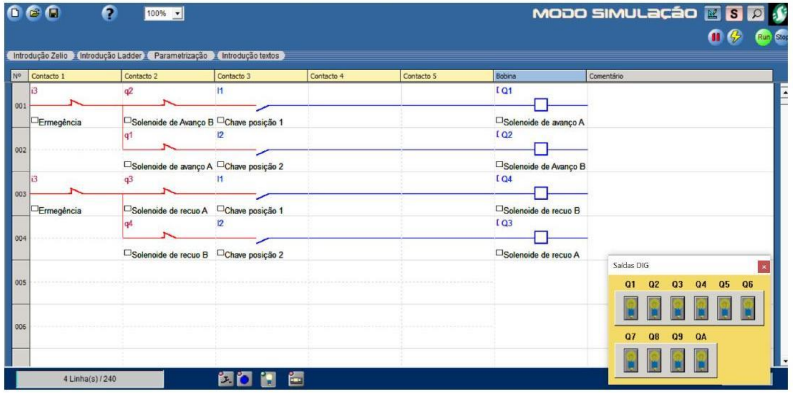
Figura 18: Diagrama da esteira



Fonte: Autoria própria, 2026

PLC Ladder Program (Zelio – Schneider)

Figura 22: Programa dos pistões



Fonte: Autoria própria, 2026

H1 = Conveyor OFF | **H2** = Conveyor ON | **H3** = Motor Overload | Interlock prevents simultaneous UP/DOWN activation

QUALITY CONTROL & PROCESSES

Two real problems identified and addressed with quality tools:

Problem 1 – PowerFlex 755 Inverter

Root Cause (5 Whys):

Lack of review of training content on storage of sensitive electronic components (ESD requirements).

PDCA Actions:

✓ ESD area implementation

✓ Training of inventory employees

✓ Review of internal procedures

✓ Temperature & humidity monitoring (20–25°C, 40–60% RH)

utilizada na Solugaraen.

Figura 27 - Folha de verificação (Bucha)

Folha de verificação

Local de coleta: Controle de qualidade — Produto: Bucha direita

Data: 23/05/2026 — Lote inspecionado: EPR2000

Responsável: Supervisor do CQ

Defeitos:	Recorrência
Deformação	4
Trinca	8
Risco na superfície	5
Diâmetro externo fora da tolerância	15
Diâmetro interno fora da tolerância	7
Outros defeitos	12
Peças aprovadas	449

Total de peças analisada: 500



Fonte: Autoria própria, 2026

Problem 2 – Bushings with Wrong Thread

Root Cause (5 Whys):

No review/validation of technical drawing with machining department before production start. 100% RH threads produced instead of 50% RH / 50% LH.

PDCA Actions:

✓ CNC program correction

✓ Detailed work instruction created

✓ Mandatory engineering–machining review

✓ Cause & effect verification sheets applied

PDCA CYCLE – CONTINUOUS IMPROVEMENT

P PLAN

Root cause analysis (5 Whys). Action plan: ESD area, employee training, CNC revision, technical drawing update.

D DO

Warehouse adapted for ESD/climate control. Employees trained. CNC programs corrected. Non-conforming bushings segregated.

PDCA

A ACT

Training added to development plan. Storage procedures standardized. Mandatory technical review before new production batches.

C CHECK

Internal audits: temperature/humidity monitoring. Dimensional inspections of machined bushings. Thread verification.

PDCA enabled Solugarden to eliminate root causes, reduce recurrence risks, and strengthen a culture of continuous improvement.

RESULTS & CONCLUSION

Productivity

Bakery Lead Time reduced by 39.2%. OEE from 62% → 80%. Monthly inventory turnover more than doubled (+117%).

Operational Costs

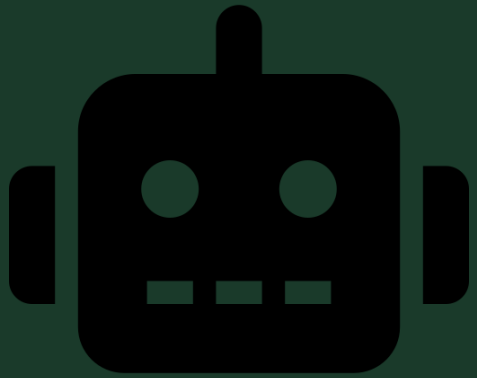
Lower corrective maintenance with elimination of hydraulic rod jamming. 30% reduction in component replacement costs.

Quality Guaranteed

PDCA + 5 Whys applied to two real problems — bushings and inverters. Standardized procedures prevent recurrence.

Validated Automation

Elevated conveyor with functional PLC and full electrical diagram for robot terrain testing at Solugarden.



THANK YOU

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